



EPBL Internship Project Report On

**Design and Verification of a Hamming Code-Based
Error Detection and Correction Encoder-Decoder
Using Verilog HDL**

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June 2026

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1 Abstract

In modern digital communication and computing systems, maintaining data integrity during transmission and storage is a critical requirement. Data transmitted through communication channels is often affected by noise, interference, signal attenuation, electromagnetic disturbances, or hardware faults, which may introduce errors into the transmitted information. Even a single-bit error can lead to incorrect processing, system failures, or data corruption in applications such as memory systems, wireless communication, satellite communication, and digital networks. Therefore, efficient error detection and correction techniques are essential to ensure reliable and accurate data communication.

This project presents the design and verification of a **Hamming (7,4) Code-Based Error Detection and Correction Encoder-Decoder** using **Verilog Hardware Description Language (HDL)**. The Hamming (7,4) code is a widely used linear block code capable of detecting and correcting single-bit errors with minimal hardware complexity. It provides an effective balance between reliability, implementation cost, and performance, making it suitable for digital communication and VLSI applications.

The proposed system consists of two major functional modules: an encoder and a decoder. The encoder accepts a 4-bit binary data input and generates a 7-bit Hamming code by calculating and inserting three parity bits at predefined positions. These parity bits provide redundancy that enables error detection and correction during data transmission. The encoded data is then considered to pass through a communication channel, where a single-bit error may occur because of external disturbances. At the receiver end, the decoder recalculates the parity values and generates a syndrome using syndrome decoding. The syndrome indicates whether an error has occurred and, if present, identifies the exact position of the erroneous bit. The decoder then corrects the error automatically and reconstructs the original 4-bit data, thereby ensuring accurate data recovery.

The complete design is developed using Verilog HDL by adopting a modular approach, where the encoder, decoder, top module, and testbench are implemented as separate modules. Functional verification is performed through simulation using industry-standard tools such as Xilinx Vivado, Xilinx ISE, or ModelSim. Various test cases are applied to validate the correctness of parity generation, error detection, and single-bit error correction. The simulation results demonstrate that the proposed design accurately identifies transmission errors, corrects single-bit faults, and successfully recovers the original transmitted data under different operating conditions.

This project enhances the understanding of digital logic design, combinational circuit implementation, parity generation, syndrome calculation, and hardware description language programming. It also provides practical exposure to simulation, verification, and debugging techniques that are widely used in VLSI design and FPGA-based system development. The implemented Hamming code architecture can be extended to more advanced error control techniques such as Single Error Correction and Double Error Detection (SECDED), BCH codes, and Reed-Solomon codes for applications requiring higher reliability. Overall, this project demonstrates an efficient, low-cost, and reliable solution for improving data integrity in digital communication systems and serves as a strong foundation for advanced studies in digital system design, VLSI, and communication engineering.

2 Introduction

Digital communication systems have become an integral part of modern technology, enabling the reliable exchange of information across computers, mobile devices, embedded systems, satellites, and high-speed networks. As the demand for faster and more efficient communication continues to grow, ensuring the accuracy and integrity of transmitted data has become increasingly important. During data transmission or storage, information may be affected by various factors such as electrical noise, electromagnetic interference, signal attenuation, hardware faults, and environmental disturbances. These factors can alter one or more bits of the transmitted data, resulting in errors that may affect the performance and reliability of the entire system.

To overcome these challenges, Error Detection and Correction (EDAC) techniques are widely used in digital communication and computer systems. These techniques enable the receiver to identify whether an error has occurred during transmission and, in many cases, automatically correct the error without requiring retransmission. The use of error correction methods improves communication reliability, reduces data loss, minimizes retransmission overhead, and enhances the overall performance of digital systems. Consequently, error correction plays a vital role in applications such as computer memory systems, wireless communication, satellite communication, optical communication, networking equipment, storage devices, and embedded systems.

Among the various error correction techniques available, the **Hamming Code** is one of the most popular and efficient linear block coding methods. Introduced by **Richard W. Hamming** in 1950, the Hamming code was developed to improve the reliability of digital communication systems by detecting and correcting errors using additional parity bits. The Hamming (7,4) code is one of the simplest forms of this technique, where four data bits are encoded into a seven-bit codeword by adding three parity bits. This coding scheme is capable of detecting and correcting any single-bit error that occurs during transmission, making it suitable for systems requiring moderate error protection with low hardware complexity.

In the Hamming (7,4) coding technique, the encoder calculates parity bits based on predefined combinations of the input data bits and combines them with the original data to form the transmitted codeword. At the receiver side, the decoder recalculates the parity values and generates a syndrome. The syndrome is a binary value that indicates whether an error has occurred and identifies the exact position of the erroneous bit. If an error is detected, the decoder automatically corrects the incorrect bit and reconstructs the original data, ensuring accurate information recovery. This process improves system reliability while maintaining low implementation cost and computational complexity.

With the advancement of Very Large Scale Integration (VLSI) technology, hardware description languages such as **Verilog HDL** have become essential tools for designing and verifying digital systems. Verilog HDL allows designers to model digital circuits at different abstraction levels, perform functional simulation, and verify circuit behavior before hardware implementation. Using Verilog, complex digital systems can be developed in a modular, scalable, and reusable manner, making it an ideal choice for FPGA and ASIC design.

In this project, a Hamming (7,4) Error Detection and Correction Encoder-Decoder is designed and implemented using Verilog HDL. The design consists of four main modules: an encoder, a decoder, a top module, and a testbench. The encoder converts a 4-bit input into a 7-bit Hamming code by generating the required parity bits. The decoder receives the encoded data, computes the syndrome, detects any single-bit error, corrects it automatically, and reconstructs the original data. The top module integrates both the encoder and decoder, while the testbench applies different input combinations and error conditions to verify the correctness of the design through simulation.

The entire design is functionally verified using simulation tools such as **ModelSim** or

****Xilinx Vivado****. Different test cases are considered to validate parity generation, error detection, and error correction. The simulation results confirm that the proposed system successfully detects and corrects single-bit transmission errors while accurately recovering the original data. This demonstrates the effectiveness of the Hamming (7,4) code in improving communication reliability.

This project provides practical knowledge of digital logic design, coding theory, parity generation, syndrome calculation, combinational logic circuits, simulation, and verification using Verilog HDL. It also strengthens the understanding of VLSI design methodologies and prepares students for advanced applications involving FPGA implementation, ASIC design, and high-reliability communication systems. Furthermore, the proposed design can be extended to advanced error correction techniques such as Single Error Correction and Double Error Detection (SEDED), BCH codes, and Reed–Solomon codes, making it a valuable foundation for future research and industrial applications in digital communication and VLSI engineering.

3 Literature Review

Error detection and correction techniques play a significant role in modern digital communication systems by ensuring the reliable transmission and storage of data. As communication technologies continue to evolve, the need for efficient error control mechanisms has increased in applications such as wireless communication, satellite communication, computer memory, networking, and embedded systems. Researchers have developed various coding techniques to detect and correct transmission errors while maintaining high system performance and low hardware complexity.

One of the earliest and most influential error correction techniques is the ****Hamming Code****, introduced by ****Richard W. Hamming**** in 1950. Hamming codes are linear block codes capable of detecting and correcting single-bit errors by adding redundant parity bits to the original data. The Hamming (7,4) code, in particular, converts four data bits into a seven-bit codeword by generating three parity bits. Because of its simple implementation and efficient performance, it has become one of the most widely used coding techniques in digital communication and computer systems.

Several studies have shown that Hamming codes provide an effective balance between hardware complexity and error correction capability. Compared to more advanced coding techniques, Hamming codes require fewer logic gates and less computational overhead, making them suitable for FPGA and ASIC implementations. Their ability to automatically identify and correct single-bit errors significantly improves communication reliability while minimizing hardware resources.

With the rapid advancement of VLSI technology, Hardware Description Languages (HDLs) such as ****Verilog HDL**** have become essential for designing and verifying digital circuits. Verilog HDL enables designers to model, simulate, test, and verify hardware systems before implementation. It supports modular design, allowing complex digital systems to be divided into smaller functional modules such as encoders, decoders, controllers, and arithmetic units. This modular approach improves design flexibility, reduces development time, and simplifies debugging.

Many researchers have implemented Hamming code encoder and decoder circuits using Verilog HDL and verified their functionality through simulation tools such as ****ModelSim**** and ****Xilinx Vivado****. Simulation results demonstrate that Hamming code-based systems successfully detect and correct single-bit transmission errors while accurately recovering the original information. These implementations are commonly used as educational examples for digital system design and as practical solutions in communication and memory applications.

Although Hamming codes are highly effective for correcting single-bit errors, they have

certain limitations. They cannot independently correct multiple-bit errors, and additional parity bits increase the length of the transmitted codeword. To overcome these limitations, researchers have proposed advanced error correction techniques such as Single Error Correction and Double Error Detection (SECDED), BCH codes, Reed–Solomon codes, Turbo codes, and Low-Density Parity-Check (LDPC) codes. These techniques provide stronger error correction capabilities but generally require greater hardware complexity, increased computational resources, and higher implementation costs.

Despite the availability of advanced coding schemes, the Hamming (7,4) code remains widely used in academic projects, FPGA prototyping, and digital system design because of its simplicity, low hardware requirements, and reliable single-bit error correction capability. It serves as an excellent foundation for understanding coding theory, parity generation, syndrome calculation, combinational logic design, and digital communication principles.

Based on the literature reviewed, implementing a Hamming (7,4) Error Detection and Correction Encoder-Decoder using Verilog HDL is an effective approach for demonstrating the principles of reliable digital communication. The project combines theoretical concepts with practical hardware design and verification, providing valuable knowledge in digital electronics, VLSI design, and communication engineering while preparing the system for future enhancements using more advanced error correction techniques.

4 Methodology

4.1 Block Diagram

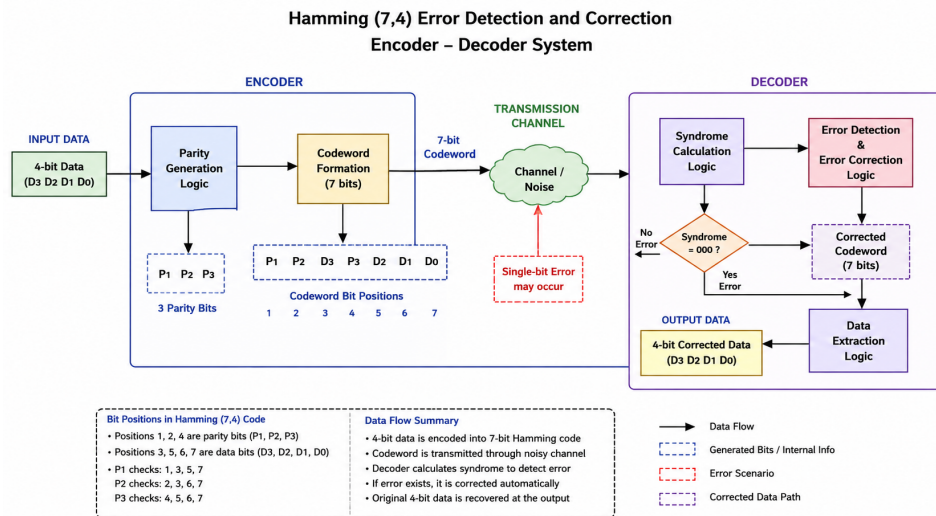


Figure 1: Block diagram

4.2 Codes

4.2.1 Encoder Design

Listing 1: Verilog Code for Hamming (7,4) Encoder

```
module encoder(
    input  [3:0] data ,
    output [6:0] code
);

assign code[2] = data[0];
assign code[4] = data[1];
assign code[5] = data[2];
assign code[6] = data[3];

assign code[0] = data[0] ^ data[1] ^ data[3];
assign code[1] = data[0] ^ data[2] ^ data[3];
assign code[3] = data[1] ^ data[2] ^ data[3];

endmodule
```

4.2.2 Decoder Design

Listing 2: Verilog Code for Hamming (7,4) Decoder

```
module decoder(
    input  [6:0] code ,
    output reg [3:0] data ,
    output reg error
);

reg [2:0] syndrome;
reg [6:0] corrected_code;

always @(*) begin

    corrected_code = code;

    syndrome[0] = code[0] ^ code[2] ^ code[4] ^ code[6];
    syndrome[1] = code[1] ^ code[2] ^ code[5] ^ code[6];
    syndrome[2] = code[3] ^ code[4] ^ code[5] ^ code[6];

    if(syndrome != 3'b000) begin
        error = 1'b1;
        corrected_code[syndrome-1] = ~corrected_code[syndrome-1];
    end
    else begin
        error = 1'b0;
    end

    data[0] = corrected_code[2];
    data[1] = corrected_code[4];

end
```

```

        data[2] = corrected_code[5];
        data[3] = corrected_code[6];

end

endmodule

```

4.2.3 Top Module

Listing 3: Top Module

```

module top_module(

    input  [3:0] data_in ,
    output [3:0] data_out ,
    output error

);

wire [6:0] encoded_data;

encoder ENC(
    .data(data_in),
    .code(encoded_data)
);

decoder DEC(
    .code(encoded_data),
    .data(data_out),
    .error(error)
);

endmodule

```

4.2.4 Testbench

Listing 4: Verilog Testbench

```

`timescale 1ns/1ps

module tb;

reg [3:0] data;
wire [6:0] code;
reg [6:0] received;
wire [3:0] decoded_data;
wire error;

encoder ENC(
    .data(data),
    .code(code)
);

```



```

decoder DEC(
    .code(received),
    .data(decoded_data),
    .error(error)
);

initial begin

    data = 4'b1011;

    #10;
    received = code;

    #10;
    received = code;
    received[4] = ~received[4];

    #10;
    received = code;
    received[0] = ~received[0];

    #10;
    received = code;
    received[6] = ~received[6];

    #10;
    $finish;

end

initial begin

$monitor("Time=%0t Data=%b Code=%b Received=%b Decoded=%b Error=%b",
    $time, data, code, received, decoded_data, error);

end

endmodule

```

5 Working Principle

The proposed system is based on the **Hamming (7,4) Error Detection and Correction** technique, which is widely used to improve the reliability of digital communication systems. The primary objective of the system is to detect and correct single-bit errors that may occur during data transmission. The complete system consists of four major modules: the **Encoder**, **Transmission Channel**, **Decoder**, and **Testbench**. Each module performs a specific function to ensure that the transmitted data is received accurately.

5.1 Input Data

The operation begins with a 4-bit binary input represented as **D3, D2, D1, and D0**. These four bits represent the original information that needs to be transmitted through the communication channel.

5.2 Encoding Process

The 4-bit input data is applied to the Hamming (7,4) encoder. The encoder generates three parity bits by performing Exclusive-OR (XOR) operations on selected combinations of the input data bits. These parity bits provide redundancy, enabling the receiver to detect and correct transmission errors.

The generated 7-bit codeword consists of four data bits and three parity bits. The bit positions are arranged as follows:

Position 7	1	2	3	4	5	6
Bit D3	P1	P2	D0	P3	D1	D2

After generating the parity bits, the encoder combines them with the original data bits to form the complete Hamming codeword.

5.3 Data Transmission

The encoded 7-bit Hamming code is transmitted through a communication channel. During transmission, the data may be affected by electrical noise, electromagnetic interference, hardware faults, or signal attenuation. These disturbances may change one of the transmitted bits, resulting in a single-bit error.

For simulation purposes, a single-bit error is intentionally introduced into the transmitted codeword using the Verilog testbench.

5.4 Syndrome Calculation

At the receiver side, the decoder receives the transmitted codeword and recalculates the parity bits. The recalculated parity values are compared with the received parity bits to generate a 3-bit syndrome.

The syndrome determines whether an error exists in the received codeword.

- Syndrome = 000 : No error is detected.
- Syndrome $\neq$ 000 : A single-bit error is detected.

The binary value of the syndrome directly indicates the position of the erroneous bit within the 7-bit codeword.

5.5 Error Detection and Correction

If the syndrome is non-zero, the decoder identifies the exact position of the erroneous bit. The incorrect bit is corrected by inverting its value (0 becomes 1 or 1 becomes 0). This automatic correction restores the original transmitted codeword without requiring retransmission.

The decoder also generates an error flag to indicate the status of transmission:

- Error = 0 : No transmission error.
- Error = 1 : Single-bit error detected and corrected.

5.6 Data Recovery

After correcting the erroneous bit, the decoder extracts the original four data bits from the corrected Hamming codeword. The recovered data matches the original input data generated by the encoder.

5.7 Functional Verification

The complete design is verified using a Verilog testbench. Different input combinations are applied, and single-bit errors are introduced at various bit positions. The simulation results demonstrate that the encoder correctly generates the Hamming code, while the decoder accurately detects the error location, corrects the erroneous bit, and reconstructs the original data.

Thus, the proposed Hamming (7,4) Encoder-Decoder provides an efficient, low-cost, and reliable solution for single-bit error detection and correction in digital communication systems. The design is implemented using Verilog HDL and can be synthesized on FPGA platforms for practical hardware applications. “latex

5.8 Software Implementation

The proposed Hamming (7,4) Error Detection and Correction Encoder-Decoder is implemented using **Verilog Hardware Description Language (HDL)**. The design follows a modular approach in which the encoder, decoder, top module, and testbench are developed as separate Verilog modules. This approach improves code readability, simplifies debugging, and allows individual modules to be tested independently before integrating the complete system.

The implementation process begins with the development of the encoder module. The encoder accepts a 4-bit binary input and generates a 7-bit Hamming code by calculating three parity bits using Exclusive-OR (XOR) operations. The generated codeword consists of four data bits and three parity bits arranged according to the Hamming (7,4) coding scheme. The encoder ensures that sufficient redundancy is added to the original data for error detection and correction.

The decoder module is then implemented using Verilog HDL. It receives the 7-bit Hamming codeword and recalculates the parity bits to generate a 3-bit syndrome. The syndrome is used to determine whether an error has occurred during transmission. If the syndrome value is zero, the received data is considered error-free. Otherwise, the syndrome indicates the exact position of the erroneous bit. The decoder automatically corrects the detected single-bit error by inverting the corresponding bit and extracts the original 4-bit data from the corrected codeword. An error flag is also generated to indicate whether an error was detected and corrected.

After implementing the encoder and decoder, a top module is created to integrate both modules into a single system. The top module establishes the connection between the encoder and decoder, allowing encoded data to flow from the transmitter to the receiver. This modular architecture improves the overall organization of the project and makes future modifications easier.

To verify the functionality of the design, a Verilog testbench is developed. The testbench applies different 4-bit input combinations to the encoder and observes the generated Hamming code. Single-bit errors are intentionally introduced at different positions in the transmitted codeword to evaluate the performance of the decoder. The simulation verifies that the decoder correctly identifies the error position using syndrome calculation, corrects the erroneous bit, and reconstructs the original data successfully.

The complete design is simulated using industry-standard EDA tools such as **Xilinx Vivado**, **Xilinx ISE**, or **ModelSim**. During simulation, waveforms are analyzed to verify parity generation, codeword formation, syndrome calculation, error detection, and data recovery. The simulation results confirm that the implemented system functions correctly under both normal and error conditions.

The software implementation demonstrates the practical application of Verilog HDL in designing reliable digital systems. It provides valuable experience in hardware modeling, functional verification, debugging, and simulation, which are essential skills in VLSI design and FPGA-based system development. The modular implementation also allows the design to be extended for more advanced error correction techniques such as Single Error Correction and Double Error Detection (SECDED), BCH codes, and Reed–Solomon codes in future work. “

6 Results and Discussion

The proposed Hamming (7,4) Error Detection and Correction Encoder-Decoder was successfully designed and implemented using Verilog HDL. The functionality of the encoder, decoder, top module, and testbench was verified through simulation using Xilinx Vivado/ModelSim. The simulation results confirmed that the system correctly performs data encoding, error detection, single-bit error correction, and recovery of the original data.

Initially, the encoder was tested by applying different 4-bit binary input combinations. For each input, the encoder generated the corresponding 7-bit Hamming code by calculating the required parity bits. The generated codewords matched the expected Hamming (7,4) encoding scheme, confirming that the parity generation logic was implemented correctly.

The decoder was then verified by applying the encoded data without introducing any transmission errors. In this case, the calculated syndrome value was equal to zero, indicating that the received codeword was error-free. The decoder successfully extracted the original 4-bit data from the received codeword, demonstrating the correct operation of the decoding process.

To evaluate the error correction capability of the system, single-bit errors were intentionally introduced at different positions in the 7-bit Hamming codeword using the Verilog testbench. During simulation, the decoder recalculated the parity bits and generated the syndrome. The syndrome correctly identified the location of the erroneous bit. The decoder then inverted the incorrect bit and reconstructed the original transmitted data successfully. The generated error flag also indicated that an error had been detected and corrected.

Several input combinations were tested to validate the reliability of the proposed design. In every test case involving a single-bit error, the decoder accurately detected the error position and restored the original data without requiring retransmission. These results demonstrate the effectiveness of the Hamming (7,4) coding technique for reliable digital communication.

The waveform obtained from the simulation clearly illustrates the complete operation of the system. The waveform shows the input data, encoded Hamming code, received codeword, syndrome generation, corrected output data, and error flag. When no error is present, the error flag remains low and the decoded output matches the input data. When a single-bit error is introduced, the error flag becomes high, the erroneous bit is corrected automatically, and the recovered output remains identical to the original input. [h] [h] The proposed design offers several advantages, including low hardware complexity, simple implementation, and reliable single-bit error correction. The modular implementation using Verilog HDL also makes the

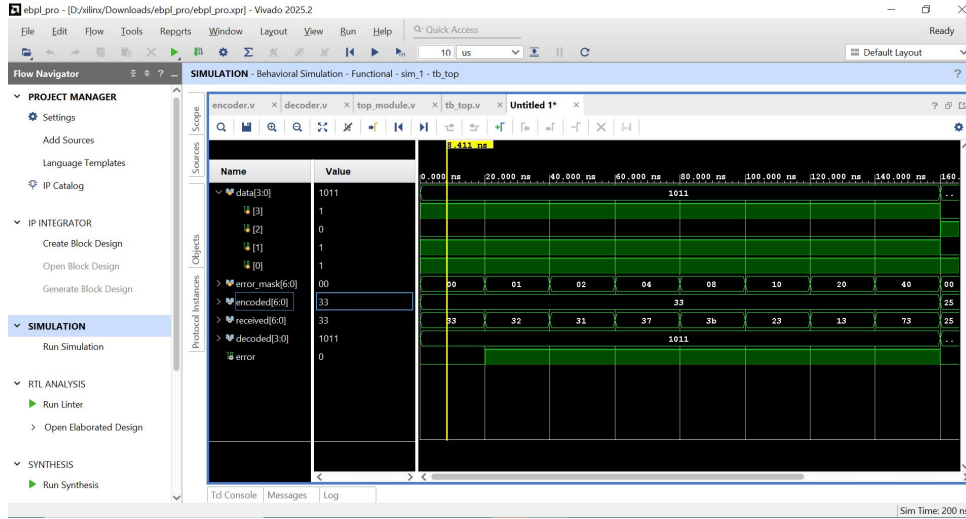


Figure 2: wave form 1

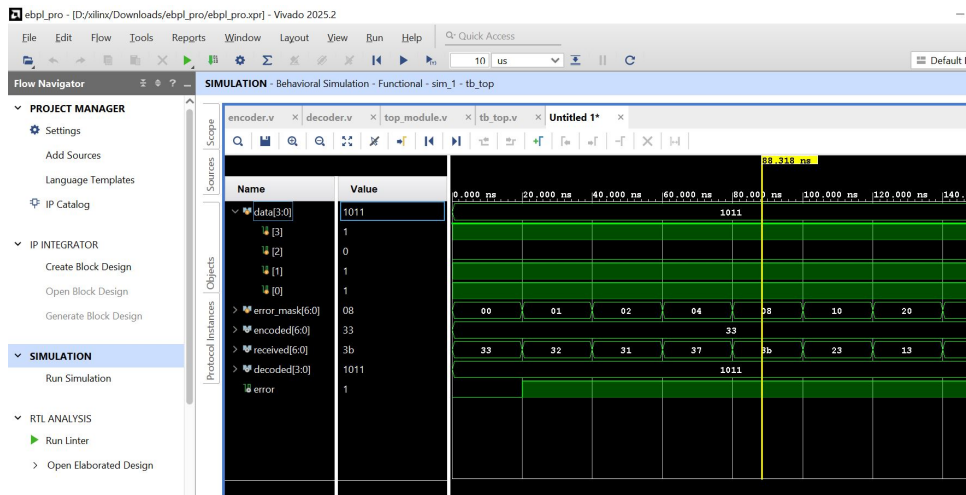


Figure 3: Wave form 2

design reusable and easy to integrate into larger digital systems. Furthermore, the design can be synthesized on FPGA platforms, making it suitable for educational, research, and industrial applications.

However, the implemented Hamming (7,4) code has certain limitations. It can correct only single-bit errors and cannot independently correct multiple-bit errors. If two or more bits are corrupted simultaneously, the decoder may not recover the original data correctly. For applications requiring higher reliability, more advanced coding techniques such as SECDED (Single Error Correction and Double Error Detection), BCH codes, or Reed–Solomon codes can be implemented.

Overall, the simulation results demonstrate that the proposed Hamming (7,4) Error Detection and Correction Encoder-Decoder performs accurately and efficiently. The successful verification of parity generation, syndrome calculation, error detection, error correction, and data recovery confirms that the proposed design satisfies the objectives of the project and provides a reliable solution for improving data integrity in digital communication systems.

7 Mapping with Program Outcomes (POs)

The proposed project on **Hamming (7,4) Error Detection and Correction Encoder-Decoder using Verilog HDL** contributes to the following Program Outcomes (POs):

PO	Contribution
PO1	Applied knowledge of digital electronics, coding theory, and communication systems to design an error correction system.
PO2	Analyzed the problem of transmission errors and selected the Hamming (7,4) coding technique as an effective solution.
PO3	Designed and implemented the encoder, decoder, top module, and testbench using Verilog HDL.
PO4	Conducted simulations and verified the functionality using FPGA design tools such as Xilinx Vivado/ModelSim.
PO5	Used modern engineering software tools for digital design, simulation, testing, and verification.
PO12	Enhanced self-learning skills by studying Hamming codes, Verilog HDL, and VLSI design methodologies.

Table 1: Program Outcome (PO) Mapping

8 Mapping with Sustainable Development Goals (SDGs)

The proposed project supports the following United Nations Sustainable Development Goals (SDGs):

SDG	Contribution
SDG 4	Quality Education – The project improves practical knowledge in digital communication, VLSI design, and hardware description languages.
SDG 9	Industry, Innovation and Infrastructure – Supports reliable digital communication systems through efficient error detection and correction techniques.
SDG 11	Sustainable Cities and Communities – Reliable communication systems contribute to smart city infrastructure and connected technologies.
SDG 12	Responsible Consumption and Production – Efficient digital designs reduce hardware complexity and optimize the use of electronic resources.

Table 2: Sustainable Development Goals (SDG) Mapping

9 Conclusion

This project successfully designed and implemented a **Hamming (7,4) Error Detection and Correction Encoder-Decoder** using **Verilog Hardware Description Language (HDL)**. The primary objective of the project was to develop a reliable digital system capable of detecting and correcting single-bit errors that may occur during data transmission. The complete system was developed using a modular approach consisting of an encoder, decoder, top module, and testbench, making the design simple, organized, and easy to verify.

The encoder accurately generated the required parity bits and produced the corresponding 7-bit Hamming codeword from a 4-bit input. The decoder successfully calculated the syndrome, detected the location of a single-bit error, corrected the erroneous bit, and recovered the original transmitted data. Functional verification through simulation confirmed that the proposed design operates correctly under both normal and error conditions. The simulation waveforms demonstrated that the system effectively maintains data integrity by automatically correcting single-bit transmission errors without requiring retransmission.

The implementation of this project provided practical knowledge of digital logic design, combinational circuit implementation, parity generation, syndrome calculation, hardware modeling, simulation, and functional verification using Verilog HDL. It also enhanced understanding of VLSI design methodologies and the use of FPGA design tools such as Xilinx Vivado and ModelSim. These skills are highly valuable for the design and verification of modern digital communication systems.

Although the Hamming (7,4) code efficiently corrects single-bit errors, it cannot independently correct multiple-bit errors. Therefore, future work may focus on implementing advanced error correction techniques such as Single Error Correction and Double Error Detection (SECDED), BCH codes, or Reed–Solomon codes to improve system reliability for more demanding communication applications.

Overall, the proposed Hamming (7,4) Error Detection and Correction Encoder-Decoder provides a simple, efficient, and reliable solution for improving data integrity in digital communication systems. The successful implementation and verification of the design demonstrate its suitability for educational purposes, FPGA-based prototyping, and as a foundation for advanced VLSI and communication engineering applications.

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